Coordinate Geometry — MCQ Worksheet

Mathematics • Chapter 7 • *25 Questions, 1 mark each — Total: 25 marks*

Instructions: *Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.*

Q1. The distance between two points (x_1, y_1) and (x_2, y_2) is:

(A) $\sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$ (B) $(x_2 - x_1) + (y_2 - y_1)$
(C) $|x_2 - x_1|$ (D) $|y_2 - y_1|$

Q2. Distance between (0, 0) and (3, 4) is:

(A) 3 (B) 4
(C) 5 (D) 7

Q3. Distance between (1, 2) and (4, 6) is:

(A) 3 (B) 4
(C) 5 (D) 6

Q4. Distance between (-3, 4) and (3, -4) is:

(A) 6 (B) 8
(C) 10 (D) 12

Q5. The midpoint of segment joining (2, 3) and (4, 7) is:

(A) (3, 5) (B) (6, 10)
(C) (2, 4) (D) (1, 2)

Q6. The midpoint of (-2, 5) and (4, -3) is:

(A) (1, 1) (B) (2, 1)
(C) (3, 2) (D) (1, 2)

Q7. Section formula: A point P divides line joining $A(x_1, y_1)$ and $B(x_2, y_2)$ internally in ratio m:n. Then P =

(A) $((mx_1+nx_2)/(m+n), (my_1+ny_2)/(m+n))$ (B) $((mx_2+nx_1)/(m+n), (my_2+ny_1)/(m+n))$
(C) $((x_1+x_2)/2, (y_1+y_2)/2)$ (D) Cannot be determined

Q8. The point dividing (3, 5) and (7, 9) in ratio 2:3 internally is:

(A) (4.6, 6.6) (B) (5.4, 7.4)
(C) (5, 7) (D) (6, 8)

Q9. If A(2, 3) and B(6, 7), midpoint M is:

(A) (2, 3) (B) (4, 5)
(C) (8, 10) (D) (6, 7)

Q10. The point on the x-axis equidistant from (5, -2) and (-3, 2) is:

(A) (0, 0) (B) (1, 0)
(C) (2, 0) (D) (-2, 0)

Q11. Area of triangle with vertices (x_1,y_1), (x_2,y_2), (x_3,y_3) is:

(A) $\frac{1}{2}|x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)|$ (B) $x_1 + x_2 + x_3$
(C) $y_1 + y_2 + y_3$ (D) Sum of distances

Q12. Three points are collinear if area of triangle formed = ?

(A) Positive (B) Negative
(C) Zero (D) Infinity

Q13. Are the points (1, 2), (2, 3), (3, 4) collinear?

(A) Yes (B) No
(C) Cannot say (D) Sometimes

Q14. Area of triangle with vertices (0,0), (4,0), (0,3) is:

(A) 6 (B) 12
(C) 7.5 (D) 3.5

Q15. The point on x-axis is of the form:

(A) (0, y) (B) (x, 0)
(C) (x, x) (D) (0, 0) only

Q16. The point on y-axis is of the form:

(A) (0, y) (B) (x, 0)
(C) (x, x) (D) (0, 0) only

Q17. Distance of point P(3, 4) from origin is:

(A) 3 (B) 4
(C) 5 (D) 7

Q18. Centroid of a triangle with vertices $A(x_1,y_1)$, $B(x_2,y_2)$, $C(x_3,y_3)$ is:

(A) $((x_1+x_2+x_3)/3, (y_1+y_2+y_3)/3)$ (B) $((x_1+x_2)/2, (y_1+y_2)/2)$
(C) $((x_2+x_3)/3, (y_2+y_3)/3)$ (D) (x_1, y_1)

Q19. The line segment joining A(2, -3) and B(5, 6) is divided by x-axis. The ratio is:

(A) 1:2 (B) 1:3
(C) 2:1 (D) 3:2

Q20. The reflection of point (3, -2) in the x-axis is:

(A) (3, 2) (B) (-3, -2)
(C) (-3, 2) (D) (3, -2)

Q21. The reflection of point (3, -2) in the y-axis is:

(A) (3, 2) (B) (-3, -2)
(C) (-3, 2) (D) (3, -2)

Q22. In which quadrant does point (-2, 3) lie?

(A) I (B) II
(C) III (D) IV

Q23. In which quadrant does (4, -5) lie?

(A) I (B) II
(C) III (D) IV

Q24. If (x, 2), (1, 3), (4, y) are collinear, value of x + y =

(A) 3 (B) 5
(C) 6 (D) 7

Q25. Length of diagonal of a square with side a (units) using distance formula is:

(A) a (B) 2a
(C) $a\sqrt{2}$ (D) $a/\sqrt{2}$

— End of Worksheet —